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Mid Term - Exploratory Data Analysis

ITAI-1371

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What We Will Accomplish

Dataset Preparation and EDA

Our group will prepare the Patient Health Records dataset for a machine learning task that predicts the Has_Disease target. We will use Python in Jupyter Notebook for every change so the original data never gets edited manually. Python will split the usable labeled records into 70% training data and 30% testing data. We will perform EDA only on the training portion and keep the testing portion separate and untouched during preprocessing development.

Cleaning and Encoding

We will clean numerical and categorical values, handle missing data with appropriate methods, and fix inconsistent formats that represent the same information. The workflow will remove identifying fields such as Patient_ID and Name because they do not describe a patient's health, while Diagnosis_Code will be removed because it could contain information too closely related to the target and create possible data leakage. Known Age inconsistencies such as twenty, -5, and 250 will be corrected and converted into numerical form, while Blood Pressure formats will be standardized, separated into systolic and diastolic values, and checked for invalid relationships. Gender, Diabetic, Smoker, Medications, and Notes will be standardized and converted into numerical features with One-Hot Encoding, while City will use frequency encoding so it remains in one numerical column. Visit dates, medications, and notes will also be prepared in consistent formats, and we will verify that the processed training features contain no remaining text or missing values before moving to the next steps.

Feature Preparation

The group will create useful numerical features from the cleaned data, including Pulse_Pressure, Mean_Arterial_Pressure, and Cholesterol_Age_Ratio. We will balance the training target so both Has_Disease classes have equal representation. Selected skewed numerical distributions will be normalized when needed, and selected continuous numerical features will be scaled to a 0-to-1 range, while One-Hot Encoded columns remain binary and the visit year and month remain in their original numerical form. Matplotlib graphs will show important before-and-after changes, and the notebook will print sample records throughout the process so each preprocessing step remains easy to follow.

Final Result

At the end of the project, we will verify the final training dataset for missing values, non-numerical features, invalid values, and class balance. We will export the cleaned training data as a CSV file that can support a later binary classification model. The completed preprocessing Jupyter Notebook will document the full process from the original training split through the final preprocessing steps. A separate Jupyter Notebook will demonstrate the differences between the training data before and after processing, including missing values, categorical encoding, class balancing, and other important changes.